

50 Guidance on prescribing

ing especially in relation to ensuring adequate fluid intake and rehydration of the affected chicks are essential in overcoming and preventing the condition. Concurrent antibacterial therapy may be required.

Complementary treatments. Recently there has been an upsurge in the marketing of complementary and herbal medicines for game birds. Very limited data are available on their efficacy and none are registered products.

Table 3 Antimicrobial doses of drugs for game birds¹

Drug	By addition to drinking water	By addition to feed
Amoxicillin	17 g/100 L 20 mg/kg body-weight	—
Apramycin	25–50 g/100 L 40 mg/kg body-weight	—
Chlortetracycline	4–12 g/100 L 30 mg/kg body-weight	200–600 g/tonne
Difloxacin	10 mg/kg body-weight	—
Enrofloxacin	5–10 g/100 L 10 mg/kg body-weight	—
Erythromycin	25.7 g/100 L	—
Lincomycin/ spectinomycin	50–150 mg of powder (Linco-Spectin)/kg body-weight	—
Neomycin	12.6 g/100 L 11 mg/kg body-weight	220 g/tonne
Sulfadiazine with trimethoprim	10.7 g/100 L 30 mg/kg body-weight	—
Sulfaquinoxaline with trimethoprim	28 g/100 L 30 mg/kg body-weight	500 g/tonne
Tetracycline	4–12 g/100 L 30–60 mg/kg body-weight	200–600 g/tonne
Tiamulin	25–30 mg/kg body-weight	200–300 g/tonne
Tilmicosin	75 mg/L 10–25 mg/kg body-weight	—
Tylosin	Treatment, 50 g/100 L, 50–200 mg/kg body-weight Prophylaxis, 12.5 g/100 L	200 g/tonne

¹drug doses for preparations that have a marketing authorisation for use in these species in the UK. Therefore unless marked ¹, the drug or doses stated are not authorised for these species

Lidocaine (human approved) is also available in 4%, 10%, and 20% preservative-free solutions for IV admixture, for direct IM administration, and premixed with D5W for IV infusion in concentrations of 2 mg/ml, 4 mg/ml, and 5 mg/ml.

Also known as lignocaine HCl. A common trade name is *Xylocaine*® (Astra).

LINCOMYCIN HCL

Prescriber Highlights -

- Lincosamide antibiotic similar to clindamycin; broad spectrum against many anaerobes, Gram + aerobic cocci, *Toxoplasma*, etc
- Contraindicated: [Horses, Rodents, Ruminants, Lagomorphs; Hypersensitive to lincosamides]
- Caution: Liver or renal dysfunction. Consider reducing dosage if severe.
- Adverse effects: gastroenteritis, pain at injection site if given IM. Rapid IV administration can cause hypotension and cardiopulmonary arrest
- Distributed into milk, may cause diarrhea in nursing animals
- Drug interactions

Chemistry - An antibiotic obtained from cultures of *Streptomyces lincolnensis*, lincomycin is available commercially as the monohydrate hydrochloride. It occurs as a white to off-white, crystalline powder that is freely soluble in water. The powder may have a faint odor and has a pK_a of 7.6. The commercially available injection has a pH of 3-5.5 and occurs as a clear to slightly yellow solution.

Storage/Stability/Compatibility - Lincomycin capsules, tablets and soluble powder should be stored at room temperature (15-30°C) in tight containers. Lincomycin injectable products should be stored at room temperature; avoid freezing.

Lincomycin HCl for injection is reportedly **physically compatible** for at least 24 hours in the following IV infusion solutions and drugs: D5W, D5W in sodium chloride 0.9%, D10W, sodium chloride 0.9%, Ringer's injection, amikacin sulfate, cephalothin sodium, chloramphenicol sodium succinate, cimetidine HCl, cytarabine, heparin sodium, penicillin G potassium/sodium (4 hours only), polymyxin B sulfate, tetracycline HCl, and vitamin B-complex with C.

Drugs that are reportedly physically incompatible when mixed with lincomycin, data conflicts, or compatibility is concentration and/or time dependent include: ampicillin sodium, carbenicillin disodium, methicillin sodium and phenytoin sodium. Compatibility is dependent upon factors such as pH, concentration, temperature and diluents used. It is suggested to consult specialized references for more specific information (e.g., *Handbook on Injectable Drugs* by Trissel; see bibliography)

Pharmacology - The lincosamide antibiotics lincomycin and clindamycin, share mechanisms of action and have similar spectrums of activity although lincomycin is usually less active against susceptible organisms. Complete cross-resistance occurs between the two drugs; at least partial cross-resistance occurs between the lincosamides and erythromycin. [They may act as bacteriostatic or bactericidal agents, depending on the concentration of the drug at the infection site and the susceptibility of the organism. The lincosamides are believed to act by binding to the 50S ribosomal subunit of susceptible bacteria, thereby inhibiting peptide bond formation.]

Most aerobic gram positive cocci are susceptible to the lincosamides (*Strep. faecalis* is not), including staphylococcus and streptococci. Other organisms that are generally susceptible include: *Corynebacterium diphtheriae*, *Nocardia asteroides*, *Erysipelothrix*, and *Mycoplasma* spp. Anaerobic bacteria that may be susceptible to the lincomycin include: *Clostridium perfringens*, *C. tetani* (not *C. difficile*), *Bacteroides* (including many strains of *B. fragilis*), *Fusobacterium*, *Peptostreptococcus*, *Actinomyces*, and *Peptococcus*.

Uses/Indications - Lincomycin has dosage forms approved for use in dogs, cats, swine, and in combination with other agents for chickens. Because clindamycin is generally better absorbed, more active, and probably less toxic, it has largely supplanted the use of lincomycin for oral and injectable therapy in small animals. Lincomycin is more economical however, and is still used by many clinicians. For further information refer to the Pharmacology or Doses sections.

Pharmacokinetics - The pharmacokinetics of lincomycin have apparently not been extensively studied in veterinary species. Unless otherwise noted, the following information applies to humans. [The drug is rapidly absorbed from the gut, but only about 30-40% of the total dose is absorbed. Food both decreases the extent and the rate of absorption. Peak serum levels are attained about 2-4 hour after oral dosing. IM administration gives peak levels about double those reached after oral dosing, and peak at about 30 minutes post injection.]

Lincomycin is distributed into most tissues. Therapeutic levels are achieved in bone, synovial fluid, bile, pleural fluid, peritoneal fluid, skin and heart muscle. CNS levels may reach 40% of those in the serum if meninges are inflamed. Lincomycin is bound from 57-72% to plasma proteins, depending on

the drug's concentration. The drug crosses the placenta and also can be distributed into milk at concentrations equal to those found in the plasma.

Lincomycin is partially metabolized in the liver. Unchanged drug and metabolites are excreted in the urine, feces and bile. Half-lives can be prolonged in patients with renal or hepatic dysfunction. The elimination half-life of lincomycin is reportedly 3-4 hours in small animals.

Contraindications/Precautions/Reproductive Safety [Although there have been case reports of parenteral administration of lincosamides to horses, cattle and sheep, the lincosamides are considered to be contraindicated for use in rabbits, hamsters, guinea pigs, horses and ruminants because of serious gastrointestinal effects that may occur, including death.]

Lincomycin is contraindicated in patients with known hypersensitivity to it or having a preexisting monilial infection.

Lincomycin crosses the placenta and cord blood concentrations are approximately 25% of those found in maternal serum. Safe use during pregnancy has not been established, but neither has the drug been implicated in causing teratogenic effects.

Because lincomycin is distributed into milk, nursing puppies or kittens of mothers taking lincomycin may develop diarrhea.

Adverse Effects/Warnings [Adverse effects reported in dogs and cats include gastroenteritis (emesis, loose stools, and infrequently bloody diarrhea in dogs).] IM injections reportedly cause pain at the injection site. Rapid intravenous administration can cause hypotension and cardiopulmonary arrest.

[Swine may also develop gastrointestinal disturbances while taking the medication.]

Overdosage/Acute Toxicity - There is little information available regarding overdoses of this drug. In dogs, oral doses of up to 300 mg/kg/day for up to one year or parenterally at 60 mg/kg/day apparently did not result in toxicity.

Drug Interactions - Kaolin [found in several over-the-counter antidiarrheal preparations) has been shown to reduce the absorption of lincomycin by up to 90% if both are given concurrently. If both drugs are necessary, separate doses by at least 2 hours.] Because lincomycin has been associated with severe diarrheas in several species, the need for an antidiarrheal product may indicate impending toxicity. Clindamycin absorption does not appear to be significantly altered by kaolin.

[Lincomycin possesses intrinsic neuromuscular blocking activity and should be used cautiously with other neuromuscular blocking agents.]

Drug/Laboratory Interactions - Slight increases in liver function tests (AST, ALT, Alk. Phosph.) may occur. There is apparently not any clinical significance associated with these increases.

Doses -

Dogs:

For susceptible infections:

- 22 - 33 mg/kg PO q12h (Aronson and Aucoin 1989)
- 15 mg/kg PO q8h; 10 mg/kg IV or IM q12h (Kirk 1989)
- 15 - 25 mg/kg PO q12h (Papich 1988)
- For mastitis: 15 mg/kg PO tid for 21 days;
For *Staph.* infections: 10 - 15 mg/kg PO tid (Morgan 1988)

Cats:

For susceptible infections:

- 22 - 33 mg/kg PO q12h (Aronson and Aucoin 1989)
- 15 mg/kg PO q8h; 10 mg/kg IV or IM q12h (Kirk 1989)
- 15 - 25 mg/kg PO q12h (Papich 1988)
- For mastitis: 15 mg/kg PO tid for 21 days;
For *Staph.* infections: 10 - 15 mg/kg PO tid (Morgan 1988)

Ferrets:

For susceptible infections:

- 10 - 15 mg/kg PO three times daily; 10 mg/kg IM twice daily (Williams 2000)

Swine:

For susceptible infections:

- For mycoplasmal (*M. hyopneumoniae*) pneumonia: Fed at 200 grams per ton of feed for 21 days or 11 mg/kg IM once daily. (Amass 1999)
- 11 mg/kg IM once daily for 3-7 days; or added to drinking water at a rate of 250 mg/gallon (average of 8.36 mg/kg/day). (Label directions; Lincocin®—Upjohn)

Monitoring Parameters -

- 1) Clinical efficacy; 2) Adverse effects; particularly severe diarrheas

Client Information - Clients should be instructed to report the incidence of severe, protracted or bloody diarrhea to the veterinarian.

Dosage Forms/Preparations/FDA Approval Status/Withholding Times -**Veterinary-Approved Products:**

Lincomycin Oral Tablets 100 mg, 200 mg, 500 mg; *Lincocin*® (Pharmacia); (Rx) Approved for use in dogs and cats.

Lincomycin Oral Solution 50 mg/ml in 20 ml dropper bottles; *Lincocin*® *Aquadrops* (Pharmacia); (Rx) Approved for use in dogs and cats.

Lincomycin Sterile Injection 100 mg/ml in 20 ml vials; *Lincocin*® (Pharmacia); (Rx) Approved for use in dogs and cats.

Lincomycin Sterile Injection 100 mg/ml in 100 ml vials; *Lincocin*® *Sterile Solution* Pharmacia & Upjohn); (OTC) Approved for use in swine Slaughter (when used as labeled) = 48 hours.

Lincomycin Sterile Injection 25 mg/ml, 100 mg/ml & 300 mg/ml in 100 ml vials. Approved for use in swine. Slaughter withdrawal (when used as labeled) = 48 hours. *Lincomix*® *Injectable* (Pharmacia) (OTC)

There are also several lincomycin combination feed/water additive products for use in swine and/or poultry.

Human-Approved Products:

Lincomycin Oral Capsules 500 mg; *Lincocin*® (Upjohn); (Rx)

Lincomycin Injection 300 mg/ml in 2 and 10 ml vials; *Lincocin*® (Upjohn), Lincomycin HCL (Steris), *Lincorex*® (Hyrex); (Rx)

LIOTHYRONINE SODIUM**Prescriber Highlights -**

- Form of T3 (active thyroid hormone) used for hypothyroidism particularly in animals not responding to T4
- Shorter duration of effect than levothyroxine
- Contraindications: acute myocardial infarction, thyrotoxicosis or untreated adrenal insufficiency. Caution: concurrent hypoadrenocorticism (treated), cardiac disease, diabetes, or elderly.
- Adverse Effects: Only associated with OD's (tachycardia, polyphagia, PU/PD, excitability, nervousness and excessive panting). Some cats may appear apathetic.
- Drug-drug; drug-lab interactions

Chemistry - A synthetically prepared sodium salt of the naturally occurring hormone T₃, liothyronine sodium occurs as an odorless, light tan crystalline powder. It is very slightly soluble in water and slightly soluble in alcohol. Each 25 micrograms of liothyronine is approximately equivalent to 60-65 mg (1 grain) of thyroglobulin or desiccated thyroid and 100 micrograms or less of levothyroxine. Liothyronine sodium may also be known as T₃, T₃ thyronine sodium, L-triiodothyronine, or Sodium L-triiodothyronine.

Storage/Stability/Compatibility - Liothyronine tablets should be stored at room temperature (15-30°C) in tight containers.

Pharmacology - Thyroid hormones affect the rate of many physiologic processes including: fat, protein and carbohydrate metabolism, increasing protein synthesis, increasing gluconeogenesis and promoting mobilization and utilization of glycogen stores. Thyroid hormones also increase oxygen consumption, body temperature, heart rate and cardiac output, blood volume, enzyme system activity, and growth and maturity. Thyroid hormone is particularly important for adequate development of the central nervous system. While the exact mechanisms how thyroid hormones exert their effects are not well understood, it is known that thyroid hormones (primarily triiodothyronine) act at the cellular level.

In humans, triiodothyronine (T₃) is the primary hormone responsible for activity. Approximately 80% of T₃ found in the peripheral tissues is derived from thyroxine (T₄) which is the principle hormone released by the thyroid.

Uses/Indications - Because of its shorter duration of action, liothyronine is generally not considered to be the drug of first choice in treating hypothyroidism. Infrequently, animals not responding to levothyroxine may respond to liothyronine.

Pharmacokinetics - In dogs, peak plasma levels of liothyronine occur 2-5 hours after oral dosing. The plasma half-life is approximately 5-6 hours. In contrast to levothyroxine, it is believed that liothyronine is nearly completely absorbed by dogs and absorption is not as affected by stomach contents, intestinal flora changes, etc.

inotropics) should be done with caution, particularly in patients with preexisting cardiomyopathy or CHF.

Laboratory Considerations - Beta blockers may produce hypoglycemia and interfere with glucose or insulin tolerance tests.

Doses -

Dogs:

- a) 0.5 - 2 mg/kg q(12)-24h (Ware 2000)
- b) 0.5 - 2 mg/kg PO q12-24h (Moise 2000)

Monitoring Parameters - 1) efficacy (ECG) 2) Adverse effects

Client Information - Relatively limited clinical experience; but appears safe. Must be dosed as prescribed. Report adverse effects to veterinarian immediately.

Dosage Forms/Preparations/FDA Approval Status/Withholding Times -

Veterinary-Approved Products: None

Human-Approved Products:

Sotalol HCl Tablets 80 mg, 120 mg, 160 mg, 240 mg; *Betapace*®—Berlex, Generic; (Rx)

SPECTINOMYCIN HCL

SPECTINOMYCIN SULFATE

Prescriber Highlights -

- Aminocyclitol antibiotic used primarily in food producing animals; relatively broad spectrum but minimal activity against anaerobes, most strains of *Pseudomonas*
- [Contraindicated: hypersensitive to it]
- Adverse effects: Appears to have minimal adverse effects at labeled dosages; probably less nephrotoxicity/ototoxicity than other aminocyclitols. Can cause neuromuscular blockade. May cause swelling at SC injection sites.
- Drug Interactions

Chemistry - An aminocyclitol antibiotic obtained from *Streptomyces spectabilis*, spectinomycin is available as the dihydrochloride pentahydrate and hexahydrate sulfate salts. It occurs as a white to pale buff, crystalline powder with pK_{as} of 7 and 8.7. It is freely soluble in water and practically insoluble in alcohol.

Storage/Stability/Compatibility - Unless otherwise instructed by the manufacturer, spectinomycin products should be stored at room temperature (15-30°C). Protect from freezing.

Pharmacology - Spectinomycin is primarily a bacteriostatic antibiotic that inhibits protein synthesis in susceptible bacteria by binding to the 30S ribosomal subunit.

Spectinomycin has activity against a wide variety of gram positive and gram negative bacteria, including *E. coli*, *Klebsiella*, *Proteus*, *Enterobacter*, *Salmonella*, *Streptococci*, *Staphylococcus* and *Mycoplasma*. It has minimal activity against anaerobes, most strains of *Pseudomonas*, *Chlamydia*, or *Treponema*.

In human medicine, spectinomycin is used principally for its activity against *Neisseria gonorrhoeae*.

Uses/Indications - Although occasionally used in dogs, cats, and horses for susceptible infections, Spectinomycin only has approved dosage forms for cattle, chickens, turkeys and swine. Refer to the Dosage section below for more information on approved uses.

Pharmacokinetics [After oral administration only about 7% of the dose is absorbed, but the drug that remains in the GI tract is active. When injected SC or IM, the drug is reportedly well absorbed with peak levels occurring in about 1 hour.]

[Tissue levels of absorbed drug are lower than those found in the serum. Spectinomycin does not appreciably enter the CSF or the eye and is not bound significantly to plasma proteins. It is unknown whether spectinomycin crosses the placenta or enters milk.]

[Absorbed drug is excreted via glomerular filtration into the urine mostly unchanged. In cattle, terminal half life is about 2 hours.]

Contraindications/Precautions/Reproductive Safety [Spectinomycin is contraindicated in patients hypersensitive to it.] The reproductive safety of the drug is not known.

Adverse Effects/Warnings - When used as labeled, adverse effects are unlikely with this drug. It is reported that parenteral use of this drug is much safer than with other aminocyclitol antibiotics, but little is known regarding prolonged use of the drug. [It is probably safe to say that spectinomycin is significantly less ototoxic and nephrotoxic than other commonly used aminocyclitol antibiotics, but can cause neuromuscular blockade.] Parenteral calcium administration will generally reverse the blockade.

[Adverse effects that have been reported in human patients receiving the drug in single or multidose studies include soreness at injection site, increases in BUN, alkaline phosphatase and SGPT, and decreases in hemoglobin, hematocrit and creatinine clearance.] Although increases in BUN and decreases in creatinine clearance and urine output have been noted, overt renal toxicity has not been demonstrated with this drug.

[Cattle receiving the sulfate form subcutaneously have developed swelling at the injection site.]

Overdosage/Acute Toxicity - No specific information was located on oral overdoses, but because the drug is negligibly absorbed after oral administration, significant toxicity is unlikely via this route.

Injected doses of 90 mg produced transient ataxia in turkey poults.

Drug Interactions [Antagonism has been reported when spectinomycin is used with chloramphenicol or tetracycline.]

Doses -

Dogs:

For susceptible infections:

- 5.5 - 11 mg/kg q12h IM or 22 mg/kg PO q12h (for enteric infections; not absorbed). (Kirk 1989)
- 5 - 10 mg/kg IM q12h. (Davis 1985)
- For acute infectious gastroenteritis: 5 - 12 mg/kg IM q12h (DeNovo 1986)

Cats:

For susceptible infections:

- For acute infectious gastroenteritis: 5 - 12 mg/kg IM q12h (DeNovo 1986)

Cattle:

For susceptible infections:

- For bronchopneumonia and fibrinous pneumonia: 33 mg/kg SC q8h. Suggested withdrawal time is 60 days. (Hjerpe 1986)
- 22 - 39.6 mg/kg/day IM divided *tid* (Upson 1988)
- For bovine respiratory disease: 10 - 15 mg/kg SC (in the neck; not more than 50 ml per site) once daily (q24h) for 3-5 consecutive days. (Label directions—*Adspec*®)

Horses:

For susceptible infections:

- 20 mg/kg IM *tid* (Robinson 1987)
- For pneumonia: 20 mg/kg IM q8h; may cause local myositis. Insufficient data to comment on use. (Beech 1987b)

Swine:

For susceptible enteric infections:

- 10 mg/kg PO q12h (Howard 1986)
- For bacterial enteritis (white scours) in baby pigs associated with *E. coli* susceptible to spectinomycin: 50 mg/10 lbs of body weight PO *bid* for 3-5 days. (Label directions; *Spectam Scour-Halt*®—Ceva)
- 10 mg/kg IM q12h (Baggot 1983)

Birds:

- For airsacculitis associated with *M. meleagridis* or chronic respiratory disease associated with *E. coli* in turkey poults (1-3 day old): Inject 0.1 ml (10 mg) SC in the base of the neck. For control and to lessen mortality due to infections from *M. synoviae*, *S. typhimurium*, *S. infantis*, and *E. coli* in newly hatched chicks: Dilute injection with normal saline to a concentration of 2.5 - 5 mg/0.2 ml and inject SC. (Label directions; *Spectam*® *Injectable*—Ceva)
- For prevention and control of chronic respiratory disease associated with *Mycoplasma gallisepticum* in broilers: Add sufficient amount to drinking water to attain a final concentration of 2 g/gallon.
For infectious synovitis associated with *Mycoplasma synoviae* in broilers: Add sufficient amount to drinking water to attain a final concentration of 1 g/gallon.
For improved weight gain/feed efficiency in floor-raised broilers: Add sufficient amount to drinking water to attain a final concentration of 0.5 g/gallon. (Label directions; *Spectam*® *Water-Soluble*—Ceva)

Monitoring Parameters -

- Clinical efficacy

Dosage Forms/Preparations/FDA Approval Status/Withholding Times**Veterinary-Approved Products:**

Spectinomycin Sulfate Injection 100 mg/ml in 500 ml vials; *Adspec*®; (Pharmacia&Upjohn); (Rx)
When used as labeled, slaughter withdrawal in cattle is 11 days; not to be used in veal calves or in dairy cattle 20 months of age or older.

Spectinomycin Injection 100 mg/ml 500 ml vials; *Spectam*® *Injectable* (Phoenix Scientific) (OTC), *Prospec*® (AgriLabs) (OTC), *Spectinomycin Hydrochloride Injectable* (Aspen, Durvet) (OTC)
Approved for use in 1-3 day old turkey poults and newly hatched chicks.

Spectinomycin Water Soluble Concentrate 0.5 g of spectinomycin per gram *Spectam*® *Water Soluble* (Bimeda); (OTC) Approved for use in chickens (not layers). Slaughter withdrawal (at labeled doses) = 5 days.

Spectinomycin Oral Solution 50 mg/ml in 240 ml pump bottle and 500 and 1000 ml without pump; *Spectam Scour-Halt*® (AgriLabs, Durvet, Phoenix Scientific); (OTC) Approved for use in swine (Weighing less than 15 lbs and not older than 4 weeks of age). Slaughter withdrawal (at labeled doses) = 21 days.

Spectinomycin combination products:

Spectinomycin/Lincomycin in a 2:1 ratio

LS 50 Water Soluble Powder (Pharmacia & Upjohn) in 2.65 oz packets, Each packet contains lincomycin 16.7 g and spectinomycin 33.3 g. Approved for use in chickens up to 7 days of age.

Lincomycin 50 mg/Spectinomycin 100 mg per ml in 20 ml vials; *Linco-Spectin*® Sterile Solution (Pharmacia & Upjohn) (OTC) Approved for use in semen extenders only. In 20 ml vials.

Human-Approved Products:

Spectinomycin Powder for Injection 400 mg (as the HCl) per ml after reconstitution. 2 g vial with 3.2 ml diluent; *Trobicin*® (Upjohn); (Rx)

SPIRONOLACTONE**Prescriber Highlights -**

- Aldosterone antagonist used as a potassium sparing diuretic
- Contraindicated: hyperkalemia, Addison's disease, anuria, acute renal failure or significant renal impairment. Caution: any renal impairment or hepatic disease
- Adverse effects: Hyperkalemia, hyponatremia and dehydration. Increased BUN and mild acidosis in patients with renal impairment. GI distress (vomiting, anorexia, etc.), CNS effects (lethargy, ataxia, headache, etc.) and endocrine changes possible.
- Drug Interactions

Chemistry - A synthetically produced aldosterone antagonist, spironolactone occurs as a cream-colored to light tan, crystalline powder with a faint mercaptan-like odor. It has a melting range of 198°-207°, with decomposition. Spironolactone is practically insoluble in water and soluble in alcohol.

Storage/Stability/Compatibility - Spironolactone tablets should be stored at room-temperature in tight, light-resistant containers. An extemporaneously prepared oral suspension can be prepared by pulverizing commercially available tablets and adding cherry syrup. This preparation is reportedly stable for at least one month when refrigerated.

Pharmacology - Aldosterone is competitively inhibited by spironolactone in the distal renal tubules with resultant increased excretion of sodium, chloride and water, and decreased excretion of potassium, ammonium, phosphate and titratable acid. Spironolactone has no effect on carbonic anhydrase or renal transport mechanisms and has its greatest effect in patients with hyperaldosteronism.

Spironolactone is not commonly used alone as most sodium is reabsorbed at the proximal tubules. Combining it with a thiazide or loop diuretic will yield maximum diuretic effect.

Uses/Indications - Spironolactone may be used in patients with congestive heart failure who do not adequately respond to furosemide and ACE inhibitors, who develop hypokalemia on other diuretics and are unwilling or unable to supplement with exogenous potassium sources. It may also be effective in treating ascites as it has less potential to increase ammonia levels than other diuretics.

Pharmacokinetics - No information was found regarding the pharmacokinetics of spironolactone in veterinary species. In humans, spironolactone is >90% bioavailable and peak levels are reached within 1-2 hours. The diuretic action of spironolactone (when used alone) is gradually attained and generally reaches its maximal effect on the third day of therapy.

Spironolactone and its active metabolite canrenone, are both about 98% bound to plasma proteins. Both spironolactone and its metabolites may cross the placenta. Canrenone has been detected in breast

LINCO-SPECTIN

賜肥金

超廣效維護猪隻健康——從出生到出售

Injection:

猪隻飼養計劃及劑量表

Injection Treatment of Linco-Spectin Every 10kg of Body Weight inject 1 cc
賜肥金注射液治療 (每10公斤體重注射1 cc)

Body Weight (Kg) 體重 (公斤)	Amount of Injection 劑量	Effective value of combination 合併效力	Remarks 附註
2	0.2 cc	30 mg	可依照劑量每 24小時注射一 次至五天為止
5	0.5 cc	75 mg	
20	2 cc	300 mg	
100	10 cc	1500 mg	

Water Soluble:

賜肥金可溶性散劑治療 (4~7天)

Amount of drinking water 每天飲水量	Berat Kg 體重 (公斤)	Amount of drug to be taken for every kg body weight 每公升 每日劑量	Remarks 附註
體重之15%	~10	7.50 mg	治療用每 200 公升 飲水加 12.5 公克 總抗生素效力
體重之12%	10~20	7.50 mg	
體重之10%	40~60	6.25 mg	
體重之8%	80以上	5.00 mg	

LINCO-SPECTIN PREMIX:

賜肥金飼料添加劑用法

Target Group (Swine) 飼料	Weight (kg) 體重 (公斤)	Amount of Linco-Spectin Premix to be add in every Tonne of food 每公噸飼料 添加公斤	Total effectiveness of antibiotic in every tonne of feed. 每公噸飼料 總抗生素效力	Remarks 附註
Newborn 乳猪	< 11kg 11以下	2.50 Kg	110 gm	As a treatment, increase the body weight and feed efficiently. 治療、增重及 增加飼料效益
Children 小猪	11~20	1.00 Kg	44 gm	
Adult 中猪	20~60	0.50 Kg	22 gm	
Old 大猪	60以上 > 60Kg	0.25 Kg	11 gm	

Indication:

適應症: Treatment and control of infectious diseases in growing swine, such as Hemolytic diarrhea, bacterial enteritis caused by Salmonella spp. and E.coli and Mycoplasma pneumoniae.

賜肥金 (lincomycin 及 spectinomycin) 適於治療革蘭氏陽性及革蘭氏陰性細菌及黴菌引起之各種疾病。

Contraindication:

禁忌:

Lincomycin is contraindicated in Rabbit, Hamster

1. Lincomycin 對兔、天竺鼠、栗鼠有毒性，故賜肥金不可用於這些類動物。
2. 與其他藥物相同，賜肥金不可用於對此兩種抗生素或其混合劑有過敏病歷之動物。

Side Effect

副作用:

For the first time treatment with this product, rarely Soft stool may occur within 36 hours. But will back to normal after 5~8 days.

偶有發現第一次以賜肥金治療36小時內有軟便發生，此種情形常在5~8天內自行消失。

Warning:

警告:

1. 賜肥金注射液應在屠宰前21天停藥。
2. 賜肥金可溶性散劑應在屠宰前8天停藥。
3. 賜肥金飼料添加劑應在屠宰前24小時停藥

For Premix powder withdrawal periods is 24 hours before slaughter

Strength

產品說明:

1. 賜肥金注射液每 cc 含 Lincomycin 50 mg, Spectinomycin 100 mg 總抗生素效力 150 mg/cc。
2. 賜肥金可溶性散劑每 100 gm 含 Lincomycin 4.2 gm Spectinomycin 8.3 gm 總抗生素效力 12.5 gm。
3. 賜肥金飼料添加劑每公升含 Lincomycin 22 gm, Spectinomycin 22 gm, 總抗生素效力 44 gm。

Packing

包裝:

1. 賜肥金注射液 (Linco-Spectin Sterile Solution) 50 cc 瓶裝
2. 賜肥金可溶性散劑 (Linco-Spectin Soluble Powder) 100 gm 瓶裝
3. 賜肥金飼料添加劑 (Linco-Spectin Premix) 5 公斤及 25 公斤裝

TUCO Division of The Upjohn Company **Upjohn**

都克 美國普強大藥廠附屬公司

台北市民生西路三巷九號N-811室 電話: 5718147, 5215571

註冊商標: 賜肥金 LINCO-SPECTIN